

PROJECT - 2 TREND STORE

Docker initialization

```
aws
Search [Alt+S]
Account ID: 7925-5750-0634
KARTHI S V

ubuntu@ip-172-31-5-189:~$ docker version

Client:
Version:      28.2.2
API version:  1.50
Go version:   gol.23.1
Git commit:   28.2.2-0ubuntu1~24.04.1
Built:        Wed Sep 10 14:16:39 2025
OS/Arch:      linux/amd64
Context:      default

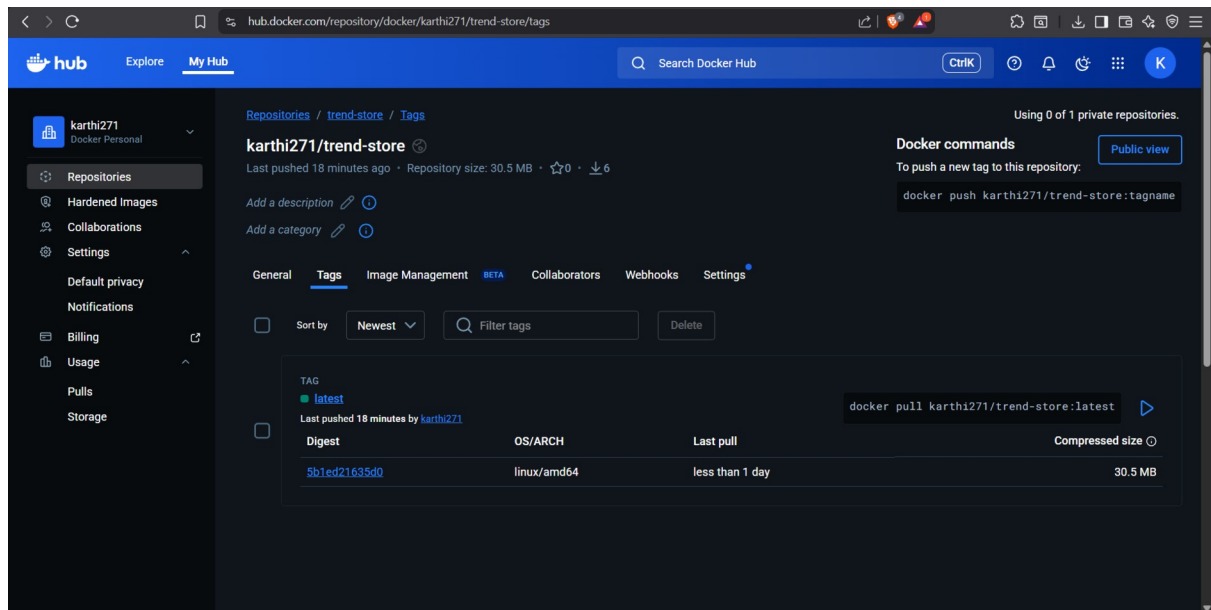
Server:
Engine:
Version:      28.2.2
API version:  1.50 (minimum version 1.24)
Go version:   gol.23.1
Git commit:   28.2.2-0ubuntu1~24.04.1
Built:        Wed Sep 10 14:16:39 2025
OS/Arch:      linux/amd64
Experimental: false
containerd:
Version:      1.7.28
GitCommit:
runc:
Version:      1.3.3-0ubuntu1~24.04.3
GitCommit:
docker-init:
Version:      0.19.0
GitCommit:
ubuntu@ip-172-31-5-189:~$
```

Running docker container

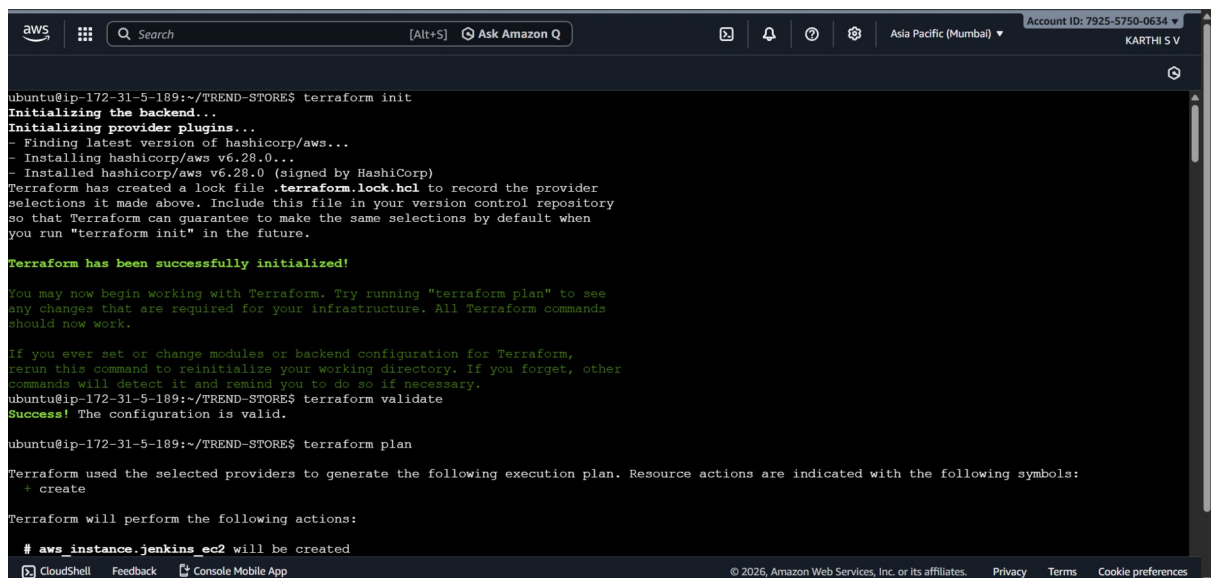
```
GUI
Launch an instance | SecurityGroup | EC2 | Trendify | Find What | EC2 Instance Co | React App CI/CD Gui | FINAL-PROJECT/Tren | KARTHI271/DevOps | +
ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressfamily=ipv4&contentType=standard&...
Account ID: 7925-5750-0634
KARTHI S V

33f95a0f3229: Download complete
33f95a0f3229: Pull complete
085c5e5aaa8e: Pull complete
0abf9e567266: Download complete
0abf9e567266: Pull complete
de54cb821236: Verifying Checksum
de54cb821236: Download complete
de54cb821236: Pull complete
Digest: sha256:8491795299c8e739b7fcc6285d531d9812ce2666e07bd3dd8db00020ad132295
Status: Downloaded newer image for nginx:alpine
--> 04da2b0513cd
Step 2/5 : RUN rm -rf /usr/share/nginx/html/*
--> Running in 36422ca72b39
--> Removed intermediate container 36422ca72b39
--> 5bb54065c202
Step 3/5 : COPY dist/ /usr/share/nginx/html/
--> 3df3edf24760
Step 4/5 : EXPOSE 3000
--> Running in ba2767a7eeb6
--> Removed intermediate container ba2767a7eeb6
--> 40946c794f3c
Step 5/5 : CMD ["nginx", "-g", "daemon off;"]
--> Running in 4d30fbc40e27
--> Removed intermediate container 4d30fbc40e27
--> 1273d3ad857e
Successfully built 1273d3ad857e
Successfully tagged trend-app:latest
ubuntu@ip-172-31-5-189:~/TREND-STORE$ docker run -d -p 3000:80 trend-app
8fc4611b4db1c399156395e88c0e3de9b583e5850a90f528174c7443ea004ce
ubuntu@ip-172-31-5-189:~/TREND-STORE$
```

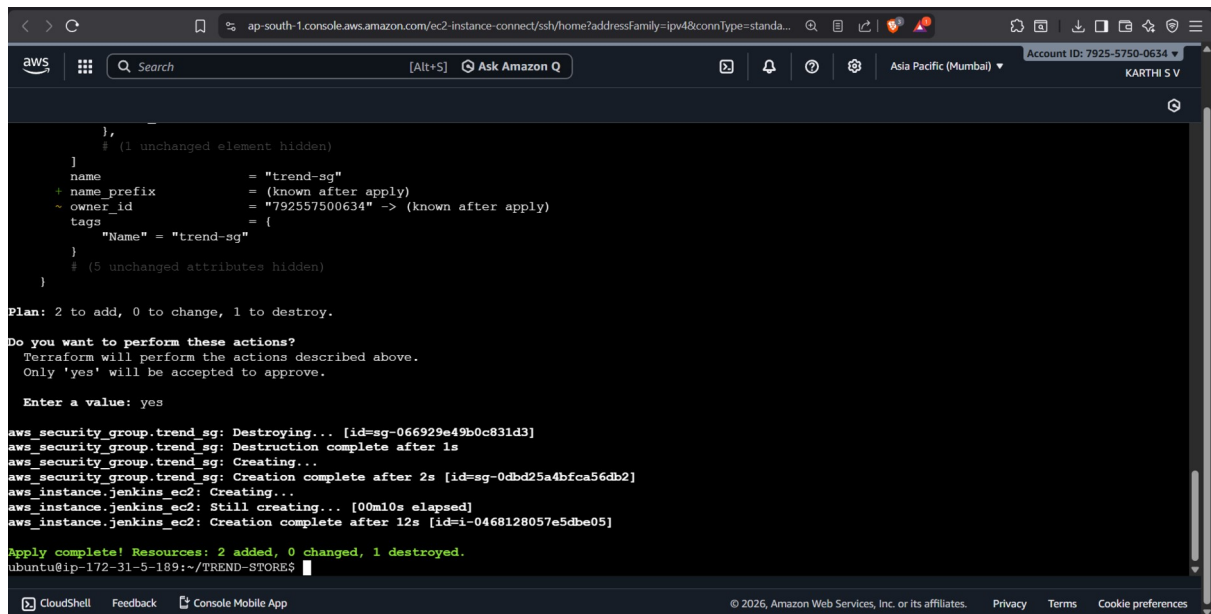
Docker hub push



Terraform



Terraform apply



The screenshot shows the AWS CloudShell interface with a terminal window. The terminal displays the output of a Terraform apply command. It shows the plan for adding two new security groups and destroying one existing one. The user confirms the actions, and the terminal shows the progress of the apply operation, including the destruction of the old security group and the creation of two new ones. The apply completes successfully, showing 2 resources added, 0 changed, and 1 destroyed.

```
},
# (1 unchanged element hidden)
}
+ name           = "trend-sg"
+ name_prefix    = (known after apply)
+ owner_id       = "792557500634" -> (known after apply)
+ tags           = {
    "Name" = "trend-sg"
  }
# (5 unchanged attributes hidden)
}

Plan: 2 to add, 0 to change, 1 to destroy.

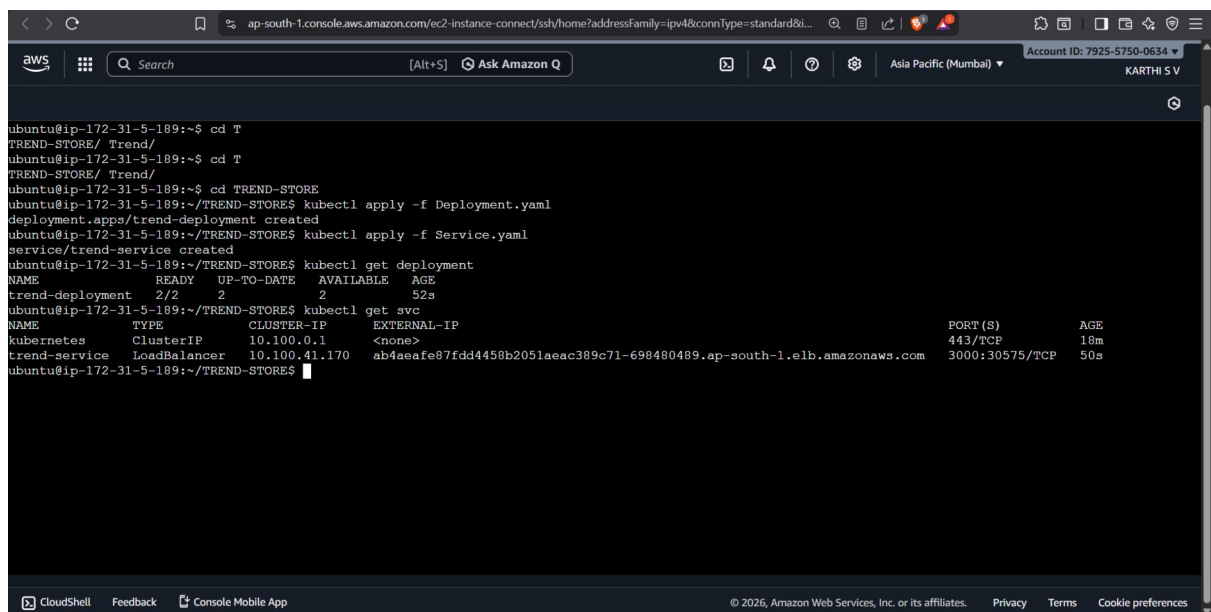
Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_security_group.trend_sg: Destroying... [id=sg-066929e49b0c831d3]
aws_security_group.trend_sg: Destruction complete after 1s
aws_security_group.trend_sg: Creating...
aws_security_group.trend_sg: Creation complete after 2s [id=sg-0dbd25a4bfca56db2]
aws_instance.jenkins_ec2: Creating...
aws_instance.jenkins_ec2: Still creating... [00m10s elapsed]
aws_instance.jenkins_ec2: Creation complete after 12s [id=i-0468128057e5dbe05]

Apply complete! Resources: 2 added, 0 changed, 1 destroyed.
ubuntu@ip-172-31-5-189:~/TREND-STORE$
```

Deployment and service



The screenshot shows the AWS CloudShell terminal with a series of commands to set up a Kubernetes deployment and service. The user navigates to the 'TREND-STORE' directory and runs 'kubectl apply' to create a deployment from 'Deployment.yaml'. Then, they run 'kubectl apply' to create a service from 'Service.yaml'. Finally, they use 'kubectl get deployment' and 'kubectl get svc' to verify the resources. The output shows the deployment is ready and the service is a LoadBalancer.

```
ubuntu@ip-172-31-5-189:~$ cd T
TREND-STORE/ Trend/
ubuntu@ip-172-31-5-189:~$ cd T
TREND-STORE/ Trend/
ubuntu@ip-172-31-5-189:~$ cd TREND-STORE
ubuntu@ip-172-31-5-189:~/TREND-STORE$ kubectl apply -f Deployment.yaml
deployment.apps/trend-deployment created
ubuntu@ip-172-31-5-189:~/TREND-STORE$ kubectl apply -f Service.yaml
service/trend-service created
ubuntu@ip-172-31-5-189:~/TREND-STORE$ kubectl get deployment
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
trend-deployment    2/2     2            2           52s
ubuntu@ip-172-31-5-189:~/TREND-STORE$ kubectl get svc
NAME      TYPE        CLUSTER-IP      EXTERNAL-IP      PORT(S)      AGE
kubernetes  ClusterIP   10.100.0.1       <none>           443/TCP      18m
trend-service  LoadBalancer 10.100.41.170    ab4aeafe87fdd4458b2051aeac389c71-698480489.ap-south-1.elb.amazonaws.com 3000:30575/TCP 50s
ubuntu@ip-172-31-5-189:~/TREND-STORE$
```

AWS client

```
ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standa...
aws [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) Account ID: 7925-5750-0634 KARTHI S V

inflating: aws/dist/awscli/customizations/wizard/wizards/lambda/new-function.yml
inflating: aws/dist/awscli/customizations/wizard/wizards/iam/new-role.yml
inflating: aws/dist/awscli/customizations/wizard/wizards/configure/_main.yml
inflating: aws/dist/awscli/customizations/wizard/wizards/events/new-rule.yml
inflating: aws/dist/awscli/customizations/sso/index.html
inflating: aws/dist/awscli/data/cli.json
inflating: aws/dist/awscli/data/metadata.json
inflating: aws/dist/awscli/data/ac.index
creating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/INSTALLER
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/top_level.txt
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/RECORD
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/METADATA
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/WHEEL
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/LICENSE
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/AUTHORS.rst
inflating: aws/dist/wheel-0.45.1.dist-info/INSTALLER
inflating: aws/dist/wheel-0.45.1.dist-info/REQUESTED
inflating: aws/dist/wheel-0.45.1.dist-info/direct_url.json
inflating: aws/dist/wheel-0.45.1.dist-info/LICENSE.txt
inflating: aws/dist/wheel-0.45.1.dist-info/METADATA
inflating: aws/dist/wheel-0.45.1.dist-info/RECORD
inflating: aws/dist/wheel-0.45.1.dist-info/entry_points.txt
inflating: aws/dist/wheel-0.45.1.dist-info/WHEEL
ubuntu@ip-172-31-5-189:~/TREND-STORE$ aws --version
Command 'aws' not found, but can be installed with:
sudo snap install aws-cli # version 1.44.14, or
sudo apt install awscli # version 2.14.6-1
See 'snap info aws-cli' for additional versions.
ubuntu@ip-172-31-5-189:~/TREND-STORE$
```

Jenkins setup

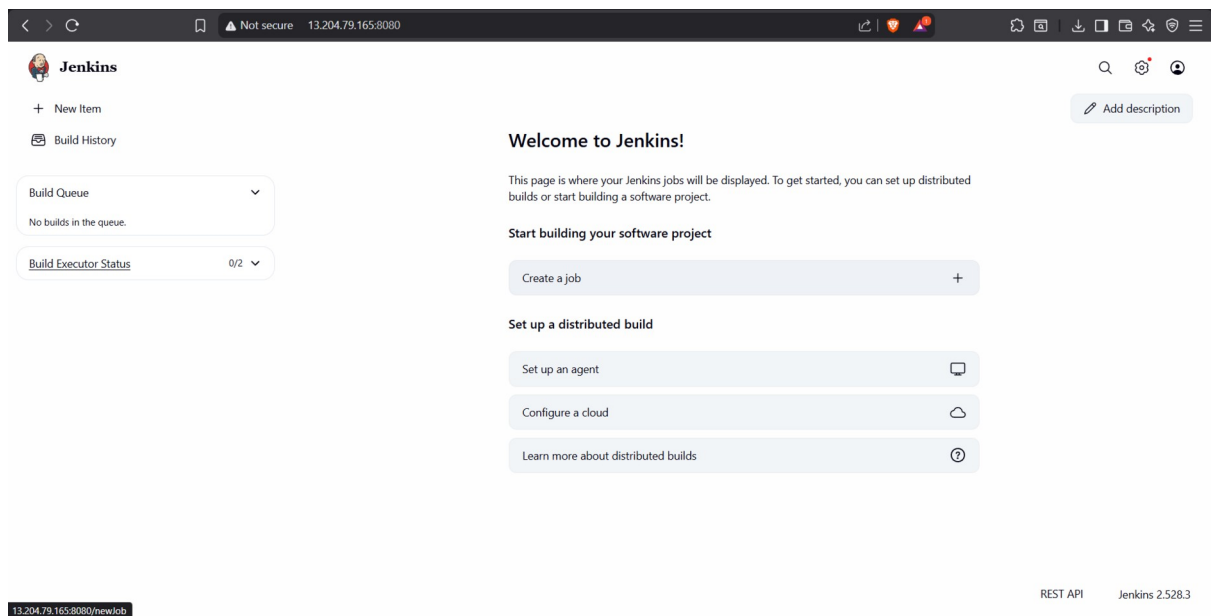
```
GUVI Launch an in SecurityGrou EC2 Inst: x React App C FINAL PROJ: KARTHI271/Fir Trendify | Fir karthi271/Are Vennilavan12 Setup Wizard: +
ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standa...
aws [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) Account ID: 7925-5750-0634 KARTHI S V

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-5-189:~$ sudo systemctl start jenkins
sudo systemctl enable jenkins
sudo systemctl status jenkins
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-01-09 16:03:09 UTC; 21s ago
     Main PID: 6941 (java)
       Tasks: 50 (limit: 2213)
      Memory: 503.9M (peak: 531.0M)
         CPU: 23.457s
    CGroup: /system.slice/jenkins.service
            └─6941 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Jan 09 16:03:02 ip-172-31-5-189 jenkins[6941]: [LF]> This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Jan 09 16:03:02 ip-172-31-5-189 jenkins[6941]: [LF]>
Jan 09 16:03:02 ip-172-31-5-189 jenkins[6941]: [LF]> *****
Jan 09 16:03:02 ip-172-31-5-189 jenkins[6941]: [LF]> *****
Jan 09 16:03:02 ip-172-31-5-189 jenkins[6941]: [LF]> *****
Jan 09 16:03:09 ip-172-31-5-189 jenkins[6941]: 2026-01-09 16:03:09.831+0000 [id=33] INFO jenkins.InitReactorRunner$1:onAttained: Comple
Jan 09 16:03:09 ip-172-31-5-189 jenkins[6941]: 2026-01-09 16:03:09.864+0000 [id=23] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is
Jan 09 16:03:09 ip-172-31-5-189 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Jan 09 16:03:11 ip-172-31-5-189 jenkins[6941]: 2026-01-09 16:03:11.722+0000 [id=49] INFO h.m.DownloadService$Downloadable#load: Obtained
Jan 09 16:03:11 ip-172-31-5-189 jenkins[6941]: 2026-01-09 16:03:11.723+0000 [id=49] INFO hudson.util.Retrier#start: Performed the action
```

Jenkins dashboard

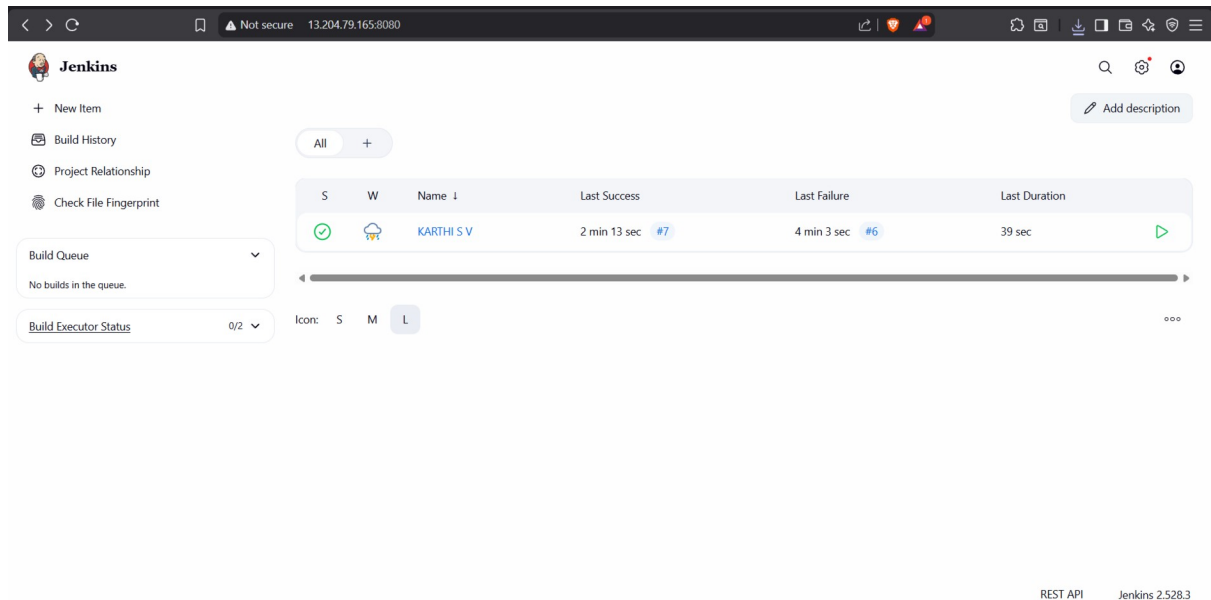


The screenshot shows the Jenkins dashboard. On the left, there's a sidebar with a 'Jenkins' logo, a '+ New Item' button, and a 'Build History' link. Below these are two dropdown menus: 'Build Queue' (showing 'No builds in the queue.') and 'Build Executor Status' (showing '0/2'). The main content area has a 'Welcome to Jenkins!' message, followed by instructions on how to get started. Below this, there's a 'Start building your software project' section with a 'Create a job' button. Further down, the 'Set up a distributed build' section includes buttons for 'Set up an agent', 'Configure a cloud', and 'Learn more about distributed builds'. At the bottom right, there are links for 'REST API' and 'Jenkins 2.528.3'.

13.204.79.165:8080/newjob

REST API Jenkins 2.528.3

Jenkins pipeline build success



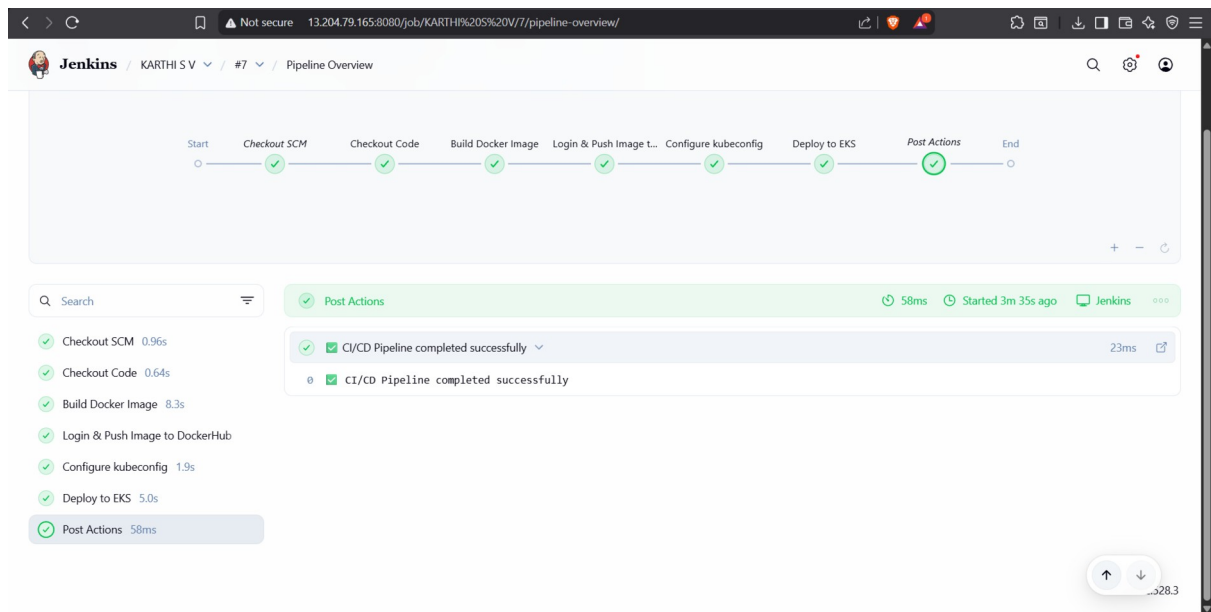
The screenshot shows the Jenkins pipeline build success page. The sidebar on the left is identical to the dashboard. The main content area features a table with columns: 'S', 'W', 'Name', 'Last Success', 'Last Failure', and 'Last Duration'. The table contains one row for a build named 'KARTHI S V'. The 'Last Success' column shows '2 min 13 sec' and '#7'. The 'Last Failure' column shows '4 min 3 sec' and '#6'. The 'Last Duration' column shows '39 sec'. Below the table, there's a 'Icon: S M L' section. At the bottom right, there are links for 'REST API' and 'Jenkins 2.528.3'.

S	W	Name	Last Success	Last Failure	Last Duration
✓	W	KARTHI S V	2 min 13 sec #7	4 min 3 sec #6	39 sec

Icon: S M L

REST API Jenkins 2.528.3

Jenkins pipeline build success



Trendify

